Computer Science Department
CS672 – Introduction to Deep Learning (CRN: 72938)
Spring 2026

Project #2 / Due 17-Apr-2026

Build a Deep Learning model, **TensorFlow** framework, (based on Neural Networks, not on classic Machine Learning models) that provides reliable and improved accuracy of a NYC Yellow Taxi **trip/ride duration**.

You used the NYC/TLC Yellow Cab dataset, in Project #1, to find correlation among the various variables to improve ride time predictions and predict the trip duration of a taxi ride considering the different factors that affect the ride duration. Along with the above-mentioned dataset, one more file should be included in your analysis, which involves the climatic conditions of the city.

Both datasets should be combined using pre-processing techniques to create a single dataset that can be used further for accurate trip duration prediction.

Get the **NYC climatic data** from the Meteostat portal.
Here is the URL to download the Jan-2020 climate data (*):
https://meteostat.net/en/place/3BBKPQ?t=2020-01-01/2020-01-31
(Note, the temperatures are Celsius degrees and not in Fahrenheit!)

NYC-Weather-Jan-2020 *

	A	B	C	D	E	F	G	H	I	J	K
1	date	tavg	tmin	tmax	prcp	snow	wdir	wspd	wpgt	pres	tsun
2	1/1/2020	3.6	1.7	5	0	0	264.6	17.3		1008.2	
3	1/2/2020	4.7	0.6	8.9	0	0	218.2	12.4		1013.9	
4	1/3/2020	7.6	6.7	8.3	2.8	0	235.5	8.4		1010.2	
5	1/4/2020	8.2	6.7	9.4	5.3	0	325.2	5.7		1003.7	
6	1/5/2020	4.6	2.8	7.2	0	0	300.1	8.2		1010.1	
7	1/6/2020	2.7	-0.5	7.8	0.8	30	239.2	16.6		1014.4	
8	1/7/2020	4.7	3.3	6.7	0	0	286	10.4		1016.2	
9	1/8/2020	2.6	-0.6	6.7	0	0	269.5	13.4		1016.6	
10	1/9/2020	-0.6	-3.3	2.2	0	0	272.3	9.1		1038.8	
11	1/10/2020	6.4	1.1	11.7	0	0	265.6	14.5		1034.5	
12	1/11/2020	14.2	8.9	21.1	0	0	192.7	19.8		1022.6	
13	1/12/2020	15.7	9.4	19.4	1	0	247.8	15.5		1018.1	
14	1/13/2020	6.1	1.7	7.8	0	0	13.4	10.4		1029	
15	1/14/2020	6.3	5	7.8	0.5	0	55	9.9		1025.3	
16	1/15/2020	8.2	6.1	11.7	0	0	269.4	9		1020.9	
17	1/16/2020	6.7	2.2	10	1.3	0	274.6	10.6		1015.1	
18	1/17/2020	-1.2	-3.9	2.2	0	0	321.2	8.8		1034.7	
19	1/18/2020	-2.2	-5	2.8	7.9	0	187.3	10.2		1028.5	
20	1/19/2020	4	0.6	7.2	0	0	269.1	10.2		1008.9	
21	1/20/2020	-1.7	-5	2.8	0	0	328.4	9.1		1024.2	
22	1/21/2020	-1.5	-4.4	3.3	0	0	323.7	8.9		1030.8	
23	1/22/2020	1	-2.2	5.6	0	0	286.8	8.1		1031.3	
24	1/23/2020	3.5	0	7.2	0	0	227.1	7.5		1029.4	

NOTE: FEEL FREE TO UTILIZE ANOTHER SOURCE FOR WEATHER-RELATED INFORMATION IN NYC!!!

Perform **regression modeling analysis** on the ready-to-be-fed data into the following Neural Network based algorithms:
1) **MLP** (Multi-Layer Perceptron)
2) **Linear Regression** (TF/Keras Sequential model w/ no hidden layers)
3) **DNN** (Deep Neural Network with at least 2 hidden layers)

Split your dataset into training and validation datasets in 80%/20% ratio.
(Note: Since dataset is time-sensitive, be extra careful on splitting…)

At the **compiling** phase, use:

- **loss function:** Mean Square Error (MSE), Mean Absolute Error (MAE).
- **optimizer**: SGD, Adam, RMSProp.
    - Use various values for **learning rate (lr)**

After each **fit** (run each one for a single value of **epoch**=100), present/plot the training_loss vs validation_loss (could utilize TensorBoard GUI module)

Present the **best Regression Deep Learning model** (compare the above 3) and its corresponding parameters. Assume bacth_size takes its default value of 32.

Using the best selected regression model, perform predictions by calling Keras **model.predict** module and review the loss.

Be aware of the following:
A) Mean square error (MSE) and mean absolute error (MAE) are common loss functions used for regression problems. MAE is less sensitive to outliers.
B) When numeric input data features have values with different ranges, each feature should be scaled independently to the same range.
C) Overfitting is a common problem for DNN models

Extra Credit

Using the **PyTorch** framework:

(a) device a neural network model with at least 2 hidden layers, like DNN you created for TensorFlow. Feel free to use the same parameters you used above.
(b) Compare and contrast the respective TensorFlow and PyTorch models.

(*) While at the **Meteostat** portal, select '**location**' (New York, Wall Street) and '**date range**' (01/01/2020 to 01/31/2020).

Note, the default 'date range' will be the current week…